TU Clausthal
Clausthal University of Technology

ETCE LAB

Limits to Growth WS 26/27
Exercise 02

Published on : **10.11.2026**
Deadline : **17.11.2026**
Do you have questions? Contact us via email etce-ltg@tu-clausthal.de

This is an individual exercise.

TASK(s) : How long does it take for the yellow and black bins to fill? What are the characteristics of your trash?

- Note the capacity and size of the bags you use for the yellow and black trash bin(s) in your house / WG / Shared kitchen.
- Note the day new trash bag is set in place (Start_Date) as well as note the day it is thrown to the container (End_Date).
- Fill out the information below and share your reflections.

Yellow bin (Gelbe Tonne)

Capacity : Liters

Days : No. of days it took to fill

Based on this information extrapolate how many liters of recyclable waste you would generate in a year. If you are sharing the trash bins, simply divide by the number of people.

Liters

Black bin (Restmüll)

Capacity : Liters

Days : No. of days it took to fill

Based on this information extrapolate how many liters of recyclable waste you would generate in a year. If you are sharing the trash bins, simply divide by the number of people.

Liters

TU Clausthal
Clausthal University of Technology

ETCE LAB

Limits to Growth WS 26/27

Occasionally observe the articles that are / were in your trash bin and try to answer following questions. Be descriptive and if you wish to add more than one observation please use semi-colon (;) in between.

Which of these articles could you have avoided purchasing in the first place?

Which of these articles that you have thrown away could have been re purposed into something more useful, hence prolonging it's life, reducing raw material extraction and waste production?

How can the product supply chain be redesigned to prevent waste (articles) from being created in the first place?

Submission Guidelines :

Please fill out the fields in this pdf upload your answers to Cloud Folder here , using password "limitstogrowth2627" as a PDF file with your name; e.g.: "E02-FirstName.pdf".

IMPORTANT (WICHTIG)
Submissions without the proper file name will NOT be considered.

Limits to Growth WS 26/27

BONUS Exercise

Choose one packaged product that you regularly use in your daily life. For example a shower gel, chips, biscuits, cosmetics etc.

Product name and Type

Now have a closer look at the package and determine the material of the packaging. Usually in Germany and in the EU, manufacturers are mandated to write the material composition of packaging and how to dispose it on the product. You'll find one or more of the following symbols on the product.



Write down the symbol and which material it indicates for the chosen product. In case of multiple materials use semicolon (;) and note down all materials.

Limits to Growth WS 26/27

If you have a weighing scale, measure the weight of the packaging material (in case of more than one materials weigh each one individually if you can separate those materials). Not down the **weight in Grams** in the following text box.

Estimate how many times in a month or in a year you buy the chosen product and calculate how much packaging waste it would be monthly and yearly in Grams.

Submission Guidelines :

Please fill out the fields in this pdf upload your answers to Cloud Folder here , using password "limitstogrowth2627" as a PDF file with your name; e.g.: "E02-FirstName.pdf"**.**

IMPORTANT (WICHTIG)
Submissions without the proper file name will NOT be considered.